

Oluwatobi Melvyn Mayungbo

Data and Machine Learning Engineer

melvyn.t@gmail.com | [LinkedIn: linkedin.com/in/oluwatobi-mayungbo-3a567026b](https://www.linkedin.com/in/oluwatobi-mayungbo-3a567026b) | [GitHub: github.com/MelvTheGoat](https://github.com/MelvTheGoat)

SUMMARY

Data and machine learning engineer who builds pipelines and checks for clean training data. Built a Prefect, dbt and DuckDB pipeline with automated data tests and drift monitoring, labelled evaluation sets with replay harnesses, and a reconciliation pipeline that sends exceptions to human review. Deployed on GCP and AWS.

SKILLS

Engineering and MLOps: Python, SQL, Prefect, dbt-core, DuckDB, Evidently, FastAPI, Docker, GCP Cloud Run, AWS ECS Fargate, PostgreSQL, Redis, Railway, MLflow, CI/CD (GitHub Actions), Git, Pytest, Streamlit, React

Data Analysis: Pandas, NumPy, SQL, Anomaly Detection

AI and LLM Systems: LLM Evaluation and Guardrails, Retrieval-Augmented Generation (RAG), Hybrid Search (BM25, Dense Embeddings, RRF), Prompt Engineering, Citation Verification, Model Context Protocol (MCP), Hugging Face, ONNX Runtime

Machine Learning: PyTorch, Scikit-learn, LightGBM, XGBoost, Deep Learning, Sequence Models (GRU, TCN, Transformers), Model Calibration (Isotonic Regression, Temperature Scaling), SHAP, Feature Engineering

Statistics and Causal Inference: Statistical Inference, Experiment Design, Power Analysis, CUPED, Causal Inference (S/T/X-Learners, Causal Forests), Uplift Modeling, Qini and AUUC, Time-Series Backtesting, Logistic Regression, WOE Scorecards

Payments and Risk: Transaction Fraud Detection, Credit Risk Decisioning, Payment Reconciliation, Double-Entry Ledgers and Idempotency, Regulatory Compliance (CBN/NDPA), Demand Forecasting, Web3 Fraud Risk Scoring

PROJECTS

Self-Operating Demand Forecasting Platform | [GitHub](#)

Tech: Python, Prefect, dbt-core, DuckDB, LightGBM, MLflow, Evidently, FastAPI, Docker, GitHub Actions

- Built an end-to-end hourly demand forecasting system for NYC taxi zones, producing 24-hour quantile forecasts through an orchestrated Prefect pipeline spanning ingestion, dbt/DuckDB transformations, backtesting, drift monitoring and conditional retraining.
- Validated with rolling-origin backtesting across 6 expanding-window folds against seasonal-naive and historical-mean baselines, reporting MASE and pinball loss, with automated leakage tests that fail the build if any feature reads past the forecast origin.

Reckon: Payment Reconciliation and Review System | [GitHub](#) | [Live App](#)

Tech: Python, PyTorch, DuckDB, FastAPI, React, Docker, Railway, PostgreSQL

- Engineered a multi-stage reconciliation pipeline using DuckDB and FastAPI, with temporal join logic and tolerance thresholds to handle asynchronous processor webhooks and bank settlement files.
- Designed a human-in-the-loop review interface in React showing match confidence scores, audit trails and suggested resolution actions to streamline exception handling.

Fintech Compliance RAG Assistant | [GitHub](#) | [Live App](#)

Tech: Python, Streamlit, BM25, ONNX Runtime, Reciprocal Rank Fusion, FastAPI, Docker, GCP Cloud Run

- Ran retrieval and generation evaluation as separate metrics to localise failure by component; traced a 0.40 refusal-correctness result to a lexical-overlap limitation in the offline stub judge and flagged re-validation against a live provider as the next step instead of overstating readiness.

web3-risk-mcp: Web3 Fraud Risk MCP Server | [GitHub](#)

Tech: Python, Model Context Protocol (MCP), Etherscan V2, GoPlus, DexScreener, Docker, GitHub Actions

- Assembled an evaluation set of 34 hand-checked addresses (12 risky, 22 safe) with a record and replay harness reporting ROC AUC, precision and recall.

EXPERIENCE

Machine Learning Instructor, SQI College of ICT

March 2026 to Present

- Teach machine learning and deep learning to student cohorts, covering statistical foundations, classical ML algorithms, neural architectures, sequence models and production evaluation metrics.
 - Instruct students on data analysis, feature engineering and model building with PyTorch, Scikit-Learn, Pandas and NumPy.
- Internal Auditor, NISER** May 2024 to July 2024
- Analysed financial and operational records to identify inconsistencies and irregularities, performing manual anomaly detection across complex transactional datasets.

EDUCATION

- B.Sc. Statistics, University of Ibadan** January 2021 to March 2025
- Professional Diploma in Artificial Intelligence, SQI College of ICT** July 2025 to Present

CERTIFICATIONS

- Google Data Analytics Professional Certificate
- Google Advanced Data Analytics Professional Certificate
- Claude Code In Action